

Database System Labs

- Get SCHEMA and Data

李旭东

LEEXUDONG@NANKAI.EDU.CN

NANKAI UNIVERSITY

OBJECTIVES

- Get the Schema of specified database In Mysql
- Get all the data of specified relation

PREPARE

- MySQL database server
- Python language
 - Access mysql

GET ALL DATABASES IN MYSQL

- show databases
- select schema_name
from **information_schema.schemata**
- select distinct table_schema
from **information_schema.tables**

	TABLE_SCHEMA
▶	information_schema
	mysql
	performance_schema
	sakila
	sys
	world

GET ALL TABLES IN MYSQL

- Way1

use world;

show tables;

```
select TABLE_NAME  
from information_schema.tables where  
table_schema='information_schema'
```

- Way2

```
select TABLE_NAME
```

```
from information_schema.tables
```

```
where table_schema='world'
```

	TABLE_NAME
▶	city
	country
	countrylanguage

GET ALL COLUMNS IN MYSQL

SELECT

COLUMN_NAME 列名, COLUMN_TYPE 数据类型,
DATA_TYPE 字段类型, CHARACTER_MAXIMUM_LENGTH 长度,
IS_NULLABLE 是否为空, COLUMN_DEFAULT 默认值,
COLUMN_COMMENT 备注

FROM INFORMATION_SCHEMA.COLUMNS

where

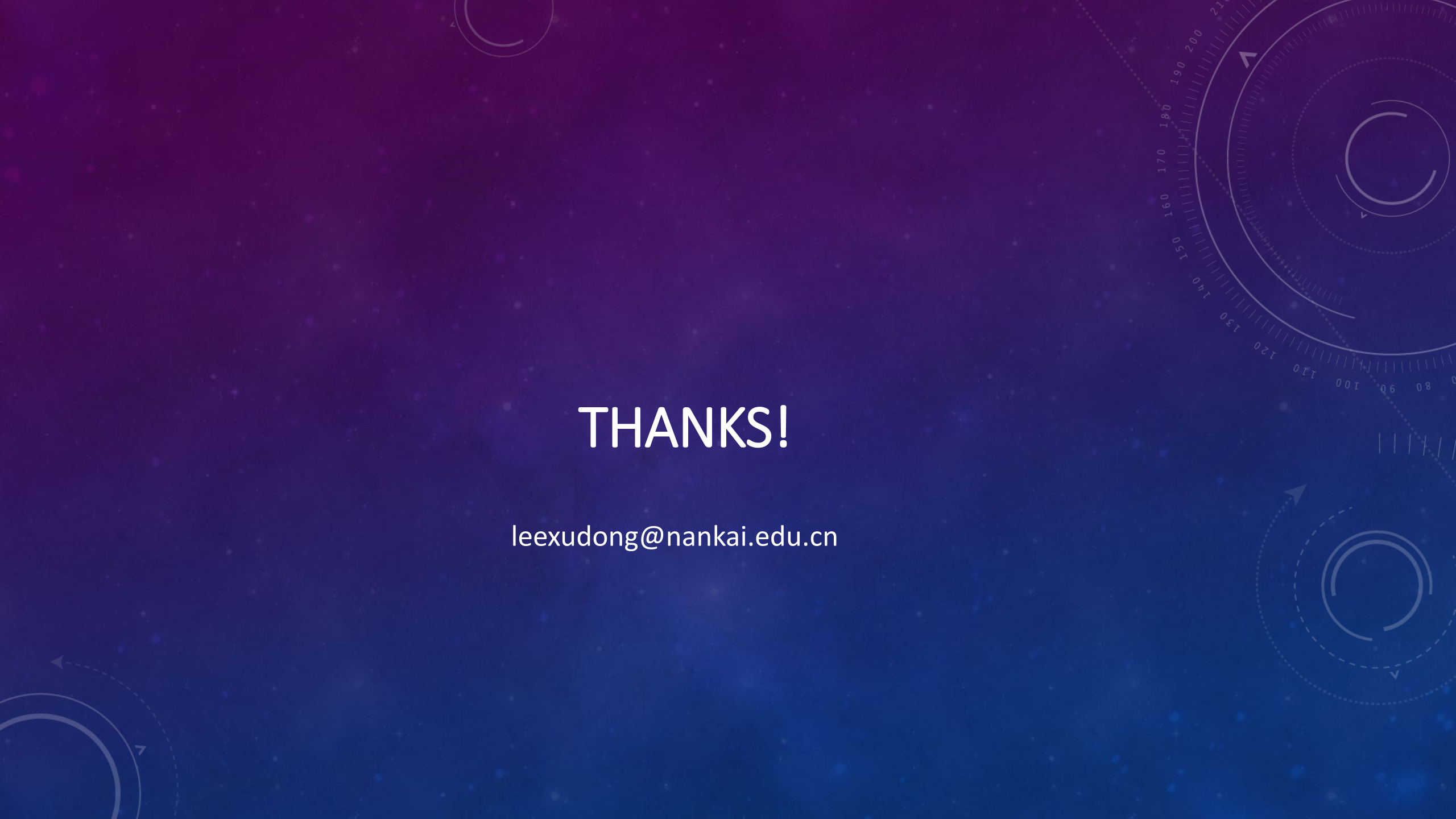
table_schema = 'world' AND table_name = 'city'

Result Grid Filter Rows: Export: Wrap Cell Content							
	列名	数据类型	字段类型	长度	是否为空	默认值	备注
▶	CountryCode	char(3)	char	3	NO		
	District	char(20)	char	20	NO		
	ID	int	int	NULL	NO	NULL	
	Name	char(35)	char	35	NO		
	Population	int	int	NULL	NO	0	

REQUIREMENTS

- 1) Write a python program to get all table schemas of university database
- 2) generate all SQL DDL of above tables
- **3) generate all SQL of insertion for one of above tables**

DO ...



THANKS!

leexudong@nankai.edu.cn